Chapter 11
The Logic of Negativity



Abstract As a logical category, negation is derived from identity as its "lying" quality, or, in accordance with Hegel's description, as the denial of non-identity. The chapter's main task, then, is to provide a dialectical critique of the positive—*formalist*—notion of Being in favor of a stratified reality consisting of the continuous differentiation of a concept with respect to (1) what it is not (*determinatio est negatio*) and to (2) all such differences made thus far in the process of knowledge. The procedural quality of knowledge, titled *Becoming*, is the main subject of the dialectical part of logic, not to be confused with or simply opposed to the notion of *Being* but rather captured as this *Being's Form* and further analyzed in terms of *Relation*. Specifically, it is the relation of the series of continuous determinations of reality to their totality, which is the *Concept*. Following Hegel, we discuss this point with respect to the problem of *infinitesimal quantities* and the notion of *Bad Infinity* that it leads to, as opposed to infinity's true form: *Finite Infinity*. In speculative hindsight, to be developed in the next chapter, this form refines the dialectical process of determination as both process and result.

Keywords Negation · Dialectical logic · Relation · Being and becoming · Infinitesimals · Bad infinity · Determinatio est negatio · Incommensurability · Existence and reality · Convergence

> And after all, where is this so different from irrational numbers—division that is never finished, a fraction of which the value will never, never, never be finally arrived at, no matter how long you may go on calculating away at it? And what can you imagine from being told that parallel lines intersect at infinity? It seems to me if one were to be over-conscientious there wouldn't be any such thing as mathematics at all.
>
> (Robert Musil, *Young Törless*)[1]

The previous chapter on formal logic and its treatment of logical concepts taught us that even in a limited approach to language, as manifested by Russell's coping with Hegel for not distinguishing between the "is" of predication and the "is" of identity,

[1] Musil (1961, pp. 97–98).

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one is forced to consider aspects of language that go well beyond a purely descriptive account of what simply and positively "is." And this concerns both predicative and identical uses of "is," which, moreover, are often difficult to separate.

First, there is a dynamic or *dialectical* aspect to them, arising from their relation to other "is"s, as in higher-order predications, number and set abstractions, and identity within contexts yet to be established. This shows us what Hegel meant when he insisted that *Being* in its pure, unqualified form must be connected to the category of *Becoming* to become *Being Proper*, or *Being-Here.* At the same time, all these differentiations take into account the specific meaning of "is," which Frege describes as part of "is a fact" or "is true," namely the *absolute* "is" of the full language—the unsurpassable frame in which all "is"-talk takes place. Ontologically, this corresponds to the concept of the *Absolute*, which is the subject of *speculative* logic.

The problem with the formal approach, thus described, is that it leaves the important logical aspects of *Being*—or the general problem of the "is" of our cognition—untouched. One aspect is precisely the inner differentiation of "is," as described above, and the other is the plurality arising from that. Both can be linked to the concepts of *Negation* and *Universality*, which stand for the respective ideas of dialectical and speculative thinking. Both are and have been a part of the formal treatment, but mainly in an implicit form, as the case of replaceability as a proxy for the universality of content has shown. Their adequate treatment, however, is yet to come, and it is their joint consideration that actually leads to a full understanding of logic proper.

11.1 Hegel's Cross

We have associated the formality of logic with the concept of identity because it represents a simplistic approach to knowledge, as captured by Nietzsche's idea that language lies by treating one thing as another without realizing it. In his words, our logic is based on these identifications and, most importantly, on treating them as natural despite their indirect, derived nature (Sect. 3.3). Our previous reflections on this led us to the concept of metaphor, which, by considering its own relational nature, takes on a triadic form and, in a second step, applies this form to itself by looking at the whole language, which makes the given "untrue" identifications a *metaphor of a metaphor* (Sect. 2.6).

I repeat all of this to show how even in the seemingly modest beginnings of identity the other logical notions are contained as part of its analysis, and with them the levels of logic mentioned in the previous chapter. Thus:

(1) There is identity itself as the original state, or some conventional beginning, which treats the *indifferences* "A = A" as given.
(2) Then comes the *denial* of this state, in the original lie of "A = B," which also leads to a relational reading of "A = A" and its use in the concept of metaphor.

(3) And finally there is the *reflection* on the previous stages, considered as all there is, from the perspective of the whole language and its identity as the metaphor of the metaphor.

This provides, in this order, the logical concepts of (1) identity, (2) negation, and (3) universality, as contained in the very act of identification as *a denial of difference* as well as in this denial's *generalizing quality*, by which many are treated as one, both of which reveal the essential quality of language as a medium of truth. Due to this, the given categories could also be thought of as representing levels of logic, namely (1) formal logic, which treats identities as forgotten lies, (2) dialectical logic, in which the lie is retrieved and employed to stand for the relationality of concepts that must always have their antipodes to be concepts at all, and (3) speculative logic, which deals with (1) and (2) as the basis for the general concept of language and its ability to lie in order to be "truer."

In light of the preceding discussion, particularly the hierarchization of identity (Chaps. 2 and 3) and the idea of reading identity in both a synchronic and diachronic way (Chap. 9), one can embed this analysis in *Hegel's Ladder*, which we linked to his exposition of the Golgotha of Spirit (Sect. 9.10), and transform it into what might be called *Hegel's Cross*. See Fig. 11.1. I deliberately leave it without additional specifications so that it can allow for multiple readings as described above.

For one, there is the relationality of the given concepts, represented by the horizontal—and synchronic—line. Here, "A = B" cannot be thought without "A = A" and *vice versa*. Then there is a vertical—and diachronic—line, standing for some origin "A = A," which as such is never absolute but always relational, both with respect to its relative specification, as given by the horizontal line stretched between the possible readings of "A = A" and "A = B" (just think of the ordinary metaphor), and with respect to its absolute specification, as given by the "A = B" below (just think of the metaphor as a metaphor). This lower "A = B" then stands for the Absolute and as such can be treated as involving all previous identities within itself.

The important point, though, is that even the absolute "A = B" *is related* to these identities, that is, is relational, yet in a specific way that determines all of them, first the relativity of "A = A" in its role of alleged beginning and its further determination as given in the horizontal line. For further details, see the section on the speculative reading of "A = B" (Sect. 3.9). Together, in rough outline, this amounts to what is known as Hegel's Golgotha of the Spirit and the ladder that was to capture its form, conceived in an upside-down direction.

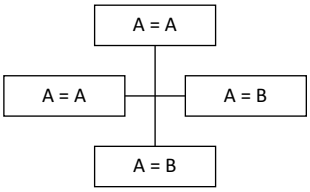


Fig. 11.1 Hegel's cross

11.2 On Leaves and Buds

What is new in this schematization of knowledge's logic by means of the Cross is not the simplification of the Ladder but the possibility of comparing it with the specific material discussed in Chap. 10, or what might be called *the Golgotha of Frege* and his formal treatment of logic. The most important aspect here, and the difference to be recognized, is the chosen beginning, which, despite its arbitrariness and particularity (or because of them), introduces a new impulse into the whole edifice. See Fig. 11.2.

In my reading, Hegel explains this aspect in the introduction to his *Science of Logic* by comparing logic to the language's grammar and the two roles it assumes. On the one hand, there is the grammar and logic taught in the textbooks, full of "dry abstractions […] that have no other value or meaning than what they immediately signify," and on the other hand there is the grammar to be felt in "the spirit and culture of a people," in which "the same rules and forms now have an enriched, living value."[2] These logics, however, do not exist on their own but always together in a mutual comparison, which is that of the particular "is" or Being and the absolute "is" or the Absolute, as Hegel's favorite example of the youth in relation to the wisdom of the old man further illustrates (Sect. 9.6):

> So logic must indeed at first be learned as something which one may well understand and penetrate into but in which, at the beginning, one misses the scope, depth, and broader significance. Only after a more profound acquaintance with the other sciences does logic rise for subjective spirit from a merely abstract universal to a universal that encompasses within itself the riches of the particular: in the same way a moral maxim does not possess in the mouth of a youngster who otherwise understands it quite well the meaning and scope that it has in the spirit of a man with a lifetime of experience, to whom therefore the weight of its content is expressed in full force.[3]

The dynamic of this relation does not present logic and its categories as something fixed; on the contrary, in a deliberate contradiction to Kant's idea of the one-and-only a priori, Hegel describes them as ever-changing, contributing to the specific self-developing form of the Spirit. This makes passages such as the following particularly important:

> […] once the substantial [logical] form of the spirit has reconstituted itself, it is of no avail to want to retain the forms of an earlier culture. These are like withered leaves pushed aside by the new buds already being generated at their roots.[4]

What Frege did, in this light, was really two interconnected things. The first was *arbitrary*, namely the choice of this or that logical vocabulary as logical or irreplaceable, which becomes particularly evident when compared with similar attempts by Bolzano, who also introduces the concepts of logical truth and entailment through irreplaceability, but without the courage to designate certain parts of language as

[2] Hegel (2010b, p. 36).

[3] Hegel (2010b, p. 37).

[4] Hegel (2010b, p. 8).

logical, since "the whole domain of concepts belonging to logic is not circumscribed so sharply that controversies could not arise at times."[5]

The other has a touch of *necessity*, seeing the "new buds" at the roots of "withered leaves," in the culture of mathematical analysis that pervades not only the mathematical physics of Newton and his reformist followers but also the everyday language inspired by it. This is what we identified as the use of multiple quantification in the complex definitions of analysis that Frege's logic allowed by linking the chain of quantifiers to placeholders of relational expressions like "*x* is smaller than *y*" or "*x* is the cause of *y*," and so on. We have mentioned this almost in passing, without emphasizing that such a possibility—namely treating relations as something logically relevant—was one of Frege's major innovations, since there is no serious precedent for it in the tradition of logic (Sect. 10.9).

This becomes even more important when we realize that one of our main expressive problems regarding the adequate concept of knowledge was precisely the relationality of the given concepts and the consequences that can be deduced from it, on which the inability to handle these relations properly must have a serious impact. In fact, one of the main failures behind the correct reading of the fallibilist doctrine as given in our four posters presented in PART I, as well as in the epistemological story behind them presented in PART II, was connected to the failure Frege analyzes in the two possible orders one can give to the commonplace expressions "for every … there is …," namely $(\forall x)(\exists y)$A and $(\exists y)(\forall x)$A, and to the relation between them.

Thus, Wittgenstein's idea that from the premise that for every failure there must be something fixed one can deduce that there is something fixed for every failure, led him to his idea of *substance* as well as to the absolute reading of the concept of unspeakability, which strongly influenced the dogmatic tone of the *Tractatus*, in contrast to its liberalization in his late writings (Sect. 4.6). The same figure is

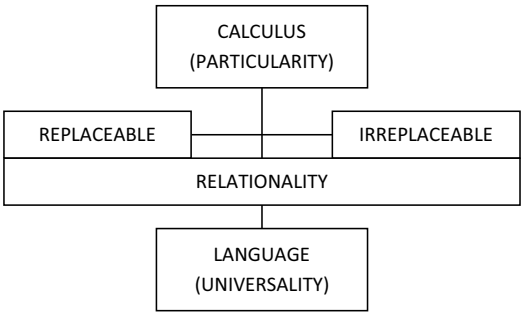


Fig. 11.2 Frege's cross

[5] Bolzano (2014, p. 59/§ 148). Bolzano's definition of logical validity otherwise closely resembles that of Frege, using the following determinations: "Let the proposition A be such that all the propositions which can be generated from it are true, if the ideas *i*, *j*... alone are considered variable, and if only objectual propositions may be formed. Then the degree of validity of A with respect to *i*, *j*... is the largest possible, i.e., it equals 1, and we can call the proposition *universally* or *fully* valid" (Bolzano 2014, p. 57/§ 147).

reflected in Descartes' belief that he could find the Archimedean point, "certain and unshakeable," by exercising the ultimate doubt first, from which it follows by a relational deduction of the given kind, and so on (Sect. 8.3). Already then, we circumscribed the difference between the grounded, groundless, and well-founded concepts of generality as pertaining to this failure in logical reasoning (Sect. 4.8), which is difficult to recognize and articulate without the logical apparatus offered by Frege. See Fig. 11.3.

On the other hand, once the given logical relations were formalized, that is, focused on and thus treated as having the particular content expressed in forms like $(\forall x)(\exists y)$A and $(\exists y)(\forall x)$A, they necessarily became limited and thus to be treated with caution, not because they are wrong, but because they became part of a larger whole—a language that continues to grow, and this also because of them. It is this phenomenon that Frege's Cross tried to capture:

(1) There is the decision to take Frege's own analysis of mathematical calculus as the beginning, in the original "A = A,"
(2) with respect to the (vertically placed) whole of language, "A = B," which Frege's analysis both represents—as a representative "part" of it—and also distorts and alters.
(3) This distortion is introduced by the horizontal line, representing the tension between the relational terms "A = A" and "A = B," indicated by the contrast between what is replaceable and what is not.

As for (2), these horizontal tensions and contradictions are described by the paradox of determination (Sect. 10.9), while the role of the paradox of universality (Sect. 10.10) lies in establishing the link between the horizontal line and the vertical line through two kinds of universality. Against this background, Frege's contribution to general logic—or to finding the new buds under the old leaves—becomes more graphical.



Fig. 11.3 Quantifiers

11.3 The Negation of Non-Identity

Once the triad of formal, dialectical, and speculative logic is phrased in terms of identity, negation, and universality, one can easily follow the advancement of these logical categories within each individual logical level, as was suggested with respect to Frege's logic. See Fig. 11.4.

This logic, in fact, contains all the given concepts as its logical constants, written in modern ideography as $=$, $\neg$, $\forall$, but with what could be called a different *potential* for breaking the formal approach and making the more adequate one appear. The concept of generality—expressed both in the paradoxes of determination and universality—has been identified as the most powerful in this respect, but such an ability is also hidden in other concepts, explicitly those of identity and negation.

In the case of identity, this disruptive potential can be easily seen by noting that both identity and generalization rely on the concept of replaceability and that, because of this link, the dialectical nature of generality can ultimately be traced back to that of identity rather than *vice versa*. Let us recall that the identity expressed by the *salva veritate* clause in (RC3) is a semantic principle essentially related to the given context. Since universal quantification, as governed by (RC4), is also a semantic principle related to objects rather than to their names, one can regard its splitting as a direct result of the contextualization of identity once it is introduced as a logical operator. In the modern philosophy of logic, it is this internal fact of Frege's logic that some of its proponents tend to express by linking existence, identity, and quantification, as Quine famously does in slogans such as "no entity without identity"[6] or "[t]o be is to be the value of a [bound] variable."[7] This gives both identity and generality a speculative—rather than dialectical—potential.

As for negativity, it is undoubtedly operative in Frege, but in two rather disconnected ways. In *Begriffsschrift* he seems to regard negation or denial (*Verneinung*) as an equivalent linguistic act to the affirmation (*Bejahung*) of propositions.[8] But he soon abandoned this, largely for the sake of simplicity, in favor of a single act, the affirmation expressed by $\vdash\!\!\!-$ and the negation becoming part of the affirmed content, as $\neg$A, rather than the affirmation itself. Much later, in a case study entitled

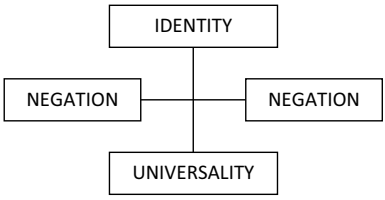


Fig. 11.4 Logic on the cross

[6] Quine (1969, p. 23, "Speaking of Objects").
[7] Quine (1963, p. 15, "On What There Is").
[8] Frege (1967, p. 13/§ 5).

"Negation," Frege gives various arguments in support of this decision, presenting it mainly as a matter of economy:

> On the assumption of two ways of judging we need 1. affirmative assertion; 2. negative assertion, e.g., inseparably attached to the world "false"; 3. a negative word like "not" in sentences uttered non-assertively. If on the other hand we assume only a single way of judging, we only need: 1. assertion; 2. a negative word. Such economy always shows that analysis has been pushed further, which leads to a clearer insight.[9]

Technically, there is nothing to complain about here, but it resembles a similar simplification that Frege adopted, namely the identification of truth with truth-value. The effect of these decisions is to gradually suppress the dual or antithetical nature of the given concepts, which, especially in the case of negation, becomes a rather pervasive feature of Frege's system. It is also a feature that Wittgenstein in his *Tractatus* was well aware of when he pointed out the redundancy of these one-sidedly introduced concepts, such as that of universal "truth-saying":

> Frege's "judgement-stroke" " ⊢ " is logically quite meaningless: in the works of Frege (and Russell) it simply indicates that these authors hold the propositions marked with this sign to be true. Thus " ⊢ " is no more a component part of a proposition than is, for instance, the proposition's number. It is quite impossible for a proposition to state that it itself is true.[10]

At the same time, Wittgenstein underestimated how negativity was implicitly operative in Frege's system, particularly through the concept of identity and its intrinsic ambivalence, which Wittgenstein thought rendered identity useless from a logical point of view.

The reason why Frege thought otherwise, and deliberately made identity an explicit part of his logical system, was precisely the strong potential of this ambivalence, or its negativity, as evidenced by the possibility of saying that in "A = B" one deliberately stipulates that two things are one. In this, identity is treated—in the spirit of Nietzsche—as a radically negative determination, or, as Hegel suggested, as a more complex concept—the denial or negation of the difference. Expressed as

"A = B" if and only if not "A ≠ B,"

it shows that "A ≠ B" actually holds by definition. Officially, in the spirit of outer positivity, Frege tried to explain away this fact by introducing the difference between sense and reference, which in turn raises problems of its own, as described above (Sect. 10.8).

Another attempt to avoid any suspicion of contradictoriness was the idea of linking identity to the Leibniz principle, which would present it as a mere consequence of our inability to differentiate between the given parts of "A = B" within the context of propositions. Since such a differentiation is always possible—simply because, by definition, A and B can always be treated as different, due to contexts such as

[9] Frege (1960, p. 130, "Negation").

[10] Wittgenstein (2001, p. 40/§ 4.442).

"I believe that A stands to the left of B"—this led, as a countermove, to a differentiation between propositions (or contexts) in which A and B (of "A = B") are replaceable *salva veritate*, as opposed to those in which they are not. Such a move, in the end, turned out to be necessary even for the sake of Frege's own ambition to introduce new objects and thus new identities, by way of stipulation, especially as expressed in the idea of abstraction (Sect. 10.7). The negativity of identity, in accordance with another of Nietzsche's insights, turned out to be the source of its creativity.

11.4 One Must Know How to Limit Oneself

All this makes the relationship between Frege and Wittgenstein and their concept of formal logic quite intricate. On the one hand, Wittgenstein underestimates the inner tensions of Frege's logical edifice, linked to the concept of predicative form, identity, and universality, and their dialectical role in the analysis of language, and tries to dissolve them within his concept of unsayability. On the other hand, he is also aware of the antithetical basis of logic that Frege was willing to sacrifice on the altar of outer positivity, considering logic as "the science of the most general laws of truth."[11]

The proper difference between their approaches is nothing other than the concept of *bipolarity*, or Wittgenstein's insight that our ability to say something always has two poles (Sect. 4.4) and that Frege's ultimate category of "is" in "is a fact" must take this into account. Let us recall that, according to Wittgenstein, to say something in a meaningful way is not only to affirm some content, but also to presuppose that it is possible for this content not to occur. The proposition is like a solid body that on the one hand restricts the freedom of movement of others but also presupposes that there is something on the other side of the boundary it sets. Tautologies and contradictions do not set such boundaries, but either leave everything open or fill it completely, thus conveying everything or nothing (Sect. 4.5). I linked these ideas to Wittgenstein's struggle against the inflation of language and the abundance of its forms (Sect. 4.1).

Although Hegel might be taken as an example of such inflationary thinking, at least on the expressive side, the general insight behind the bipolarity of propositional form is something that he undoubtedly shares with Wittgenstein, namely that "if I want to say something *meaningful*, I have to limit myself." One of the readings that can be given to this on a colloquial level is that there are *too many* topics that can be talked about, and that to say something means to select some of them in opposition to others:

> Someone who wants to do something great must know, as Goethe says, how to limit himself. By contrast, someone who wants everything in fact wants nothing and accomplishes nothing. There are a lot of interesting things in the world: Spanish poetry, chemistry, politics, music. All of that is very interesting, and one cannot blame anybody who takes an interest in them.

[11] Frege (1979, p. 128, "Logic").

> However, if as an individual one wants to achieve something in a particular situation, one must stick to something determinate and not split up one's power in various directions.[12]

The advantage of this formulation is that it naturally associates negation with universality as a means of its determination, and as such makes negativity an integral part of any piece of knowledge that can be acquired. The decision to choose this rather than that is inherent in the content of knowledge, not something that can be added to it.

Hegel first translates this connection into epistemological terms, significantly in his *Phenomenology*, where he considers the possibility of obtaining knowledge based on sensuous-certainty to be the most concrete that there is (Sect. 3.3). This utmost particularity, however, turns out to be the ultimate form of universality, incapable of making any substantial difference. In order to arrive at some knowledge, I must be able to abstract from the given moment, from "Here" and "Now," and thus lie, according to the nature of language, but in a way that is truer or particular enough to limit the utmost universality of the use of "Here"s and "Now"s in every possible situation (Sect. 3.4). This is what the identity of "A = B" stands for, as opposed to its tautological sister "A = A."

In the *Science of Logic*, the logical phrasing of the same comes to the fore by self-reflectively basing the entire edifice on a systematic determination of the yet-undetermined—or too general—categories, by means of negation. This determination begins with the concept of *Pure Being*, which we already read as the unqualified "is," or being "without further determination […], equal only to itself and also not unequal with respect to another."[13] Since such a lack of differentiation allows us to say nothing, Hegel interprets it as identical with *Pure Nothing*, which is as good as an unqualified denial, or "is not." "Nothing is therefore the same determination or rather absence of determination, and thus altogether the same as what pure *being* is."[14]

Hegel's reference to Goethe—or rather to a passage from his poem "Natur und Kunst," namely that "in der Beschränkung zeigt sich erst der Meister"[15]—proves to be important here, developed from another perspective as a reference to Goethe's theory of colors. For Goethe, pace Newton, colors—and, by extension, everything that can be seen—are not positive determinations, such as those given by the angle of a ray's refraction, but rather phenomena arising *from the contrast* between light and darkness:

> What we have been describing is an archetypal phenomenon of this kind. On the one hand we see light or a bright object, on the other, darkness or a dark object. Between them we place turbidity and through this mediation colors arise from the opposites […]."[16]

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[12] Hegel (2010a, p. 127/§ 80).

[13] Hegel (2010b, p. 59).

[14] Hegel (2010b, p. 59).

[15] Goethe (2005, pp. 124–125): "Mastery is revealed in limitation [...]."

[16] Goethe (1988, p. 195/§ 175, "Theory of Color, Didactic Section").

Hegel translated this insight quite straightforwardly into the general conditions of logic:

> Pure light and pure darkness are two voids that amount to the same thing. Only in determinate light (and light is determined through darkness: in clouded light therefore), just as only in determinate darkness (and darkness is determined through light: in illuminated darkness therefore), can something be distinguished, since only clouded light and illuminated darkness have distinction in them and hence are determinate being, *existence*.[17]

To see or say something one must limit oneself by establishing a contrast rather than simply referring to a positively given object. See Fig. 11.5. Such a reference is arguably possible only against the background of all that "is," or "is not" there, by its further determination, which is mutual. But this mutuality, one might say, is governed by the idea of negativity, as a limitation of generality, along the lines of what Hegel expresses with Spinoza's dictum: *omnis determinatio est negatio.*

The idea of determination, which is basically a negative step, is one of Hegel's main trademarks as well as the point at which the dialectical logic diverges from its formal variant. That it was already operative there, in the very principle of determination, was discussed in the previous chapter. The lesson to be drawn from this is that the problem of formalism lies not in the distinctions it makes but in the attitude it adopts towards them, which is a positivity unaware of, or alienated from, its negative component. The main merit of formalism, then, and the contribution it makes to the whole enterprise, is that it is bold enough to start somewhere—that is, to take the step of treating some aspects of language as logical, as Frege did in contrast to Bolzano. The main hitch of formalism is that it treats itself as something fixed and inevitable.

11.5 The Formalism of Being

It is not surprising that it is Descartes who attacks the formal logic of his scholastic predecessors as conventional,[18] to be replaced by his method of "clare et distincte," simply because he makes the same mistake as they did. But, as Peirce remarks, formal

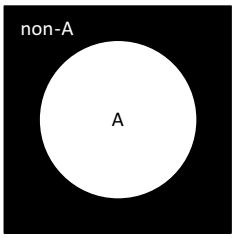


Fig. 11.5 Light and darkness

[17] Hegel (2010b, p. 69).

[18] See, for example, Descartes (2006, pp. 16–17).

logicians are better off than Descartes because they at least do not deny that there is the possibility of other beginnings as well (Sect. 8.3) and, what is more, they show that their schematic beginning is making substantial progress in many disciplines, as Leibniz and Newton were making in their introduction of calculus; something which, by the way, Descartes made a substantial use of in his analytic geometry. The problem of formalism, then, is not its subject—which is the negativity of knowledge as based on the schematism's lying—but rather its attitude toward it.

This becomes even more obvious once we see that not only does Hegel himself employ formalist thinking, as in his substantivization and the language of identities "A = A" and "A = B" with which he systematically works, but that he, pace Wittgenstein, does not believe that this can be done in a self-explanatory way, without errors and ambivalences, which are, as he knows, not only vices of language but also its most valuable virtues. The identification of Being with Nothingness is, obviously, the most formal and absurd claim one can make, equivalent to what Wittgenstein calls the philosophical rubbish in sentences like "whether the good is more or less identical than the beautiful."[19] Again, it is not surprising that Russell, in his introduction to the *Tractatus*, chooses this as the paragraph with which he can most identify,[20] without noticing that, ironically, it is no different from Wittgenstein's own claim about the nonsense of tautologies and contradictories, without the unnecessary hysteria of unsayability about it.

Hegel has his own way of dealing with the formality of propositions such as "Being is Nothing," namely that they are in fact part of a speculative proposition that, in order to be properly understood, cannot be read *one-sidedly*, but as a proposition that is "*just as false as it is correct*."[21] And this is precisely the expression of the given concepts' bipolarity, or, more precisely, the recognition that there is a space in which determinate content is expressed as something that makes a difference, which means that all "is" is affected by "is not" and *vice versa*. The advantage over Wittgenstein's terminology is that it leaves these expressions within the framework of language and its logic, in its higher-level stratification of formality, dialecticity, and speculativity.

The key moment in this metalogical differentiation is the relationality of Being and Nothing, which can be transformed into what it is not, namely the absolute reading of relationality, as discussed above. Hegel confronts this problem, or what he calls the inseparatedness of being and nothing,[22] by introducing the concept of *Becoming* (*Werden*), described as the moment when one ceases to be while the other comes into being:

> Becoming is a ceaseless unrest that collapses into a quiescent result. This can also be expressed thus: becoming is the vanishing of being into nothing, and of nothing into being, and the vanishing of being and nothing in general; but at the same time it rests on their

[19] Wittgenstein (2001, p. 23/§ 4.003).

[20] Wittgenstein (2001, p. xi).

[21] Hegel (2010b, p. 67).

[22] Hegel (2010b, p. 80).

> being distinct. It therefore contradicts itself in itself, because what it unites within itself is self-opposed; but such a union destroys itself.[23]

It is advisable to read this not as new, more real concept of Being to be studied by dialectical logic itself, as proposed by Marx (or rather Engels), but as a reflection on the *relationality* of our concepts and their expression within the system, starting from the very opposition of "is" and "is not." This is only confirmed by Hegel's establishment of the connection between Becoming and the core of the dialectical method, that is, the sublation of the antithetical sides into a new thesis representing a new kind of Being, or rather the transformation of the old concept of it.[24]

Grammatically speaking, one can follow Stekeler here and read the category of Becoming in terms of the German "wird,"[25] but in fact in terms of all the grammatical categories that resist the mere formalism of the simple indicative mood, such as "was," "will be," "would be," and so on. They all stand for something transitory, or, put differently, they serve to differentiate more finely within what "is" and "is not," emphasizing their relationality—that is, the constant need for specification—and the process that underlies them. *Pure Being* is, in this sense, the abstract form of the Absolute—or the general "is" of knowledge—that is behind the whole process and, in a sense, underlies the development of something already present, transforming Hegel's cross or ladder into a circle. See Fig. 11.6.

In this light, the category of Becoming within the edifice of formal logic represents the *instability* of our knowledge, and as such does not correspond to a specific *force* in nature, nor to a transient state that moves us and could be studied independently of Being. Rather, it stands for the relationality of our distinctions, which are distinctions or differences rather than positive facts. Becoming belongs to Being in its general form, arising from the overall negativity that affects this Being, and must be treated as such for the sake of this negativity.

Such a negativity itself has a peculiar structure, representing not only the difference between two poles to be established, but also one of these poles itself, thus

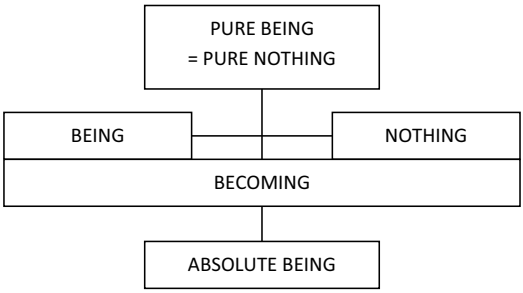


Fig. 11.6 Determination of categories

[23] Hegel (2010b, p. 81).
[24] Hegel (2010b, pp. 81–82).
[25] Stekeler (2019, p. 142).

exemplifying what we now call "X = X + Y." Its specific feature, then, could be described by the link to generality, or by negation working as its inner determination—or as a limitation by a relative split. Thus, in contrast to the previous exegesis, which seemed to begin with something particular, such as a picture of *pavimento*, and to extend it with something else, namely, a picture of *pavimento*'s linear projection (Sect. 6.8), here one can see the progress as proceeding in the opposite direction, beginning with the most general concept there is, namely that of Pure Being—which, as we know now, is identical with Pure Nothing—in order to make it more particular.

This is, in fact, how the concept of particular existence or *Being-Here* (*Dasein*) arises, which is described as a product of Becoming in its triadic structure:

> It is not mere being but *existence*, or *Dasein*; according to its etymology, it is being [*Sein*] in a certain *place* [*da*]. But the representation of space does not belong here. As it follows upon becoming, existence is in general *being* with a *non-being*, so that this non-being is taken up into simple unity with being. *Non-being* thus taken up into being with the result that the concrete whole is in the form of being, of immediacy, constitutes *determinateness* as such.[26]

At this moment a fixed boundary is drawn between what some A is and what it is not, thus establishing negation in the proper sense of the word. It is this kind of negation that Hegel calls "determinate"; that is, one that leads to a fixed or "finite" determination rather than a mere denial that something is not what we thought it was, as linked with the transient nature of our experience captured by the concept of Becoming.

11.6 Infinitesimal Quanta

The differences between *Being*, *Becoming*, and *Being-Here* could easily become an empty play on words, and one can agree with Stekeler that there is no clear sense *per se* of what Hegel's phrases, such as "becoming is a ceaseless unrest that collapses into a quiescent result," can mean.[27]

Fortunately, Hegel provides many interpretive keys (perhaps too many), the most appropriate for our purposes being that of the *infinitesimal quantum*. This concept itself goes back to Newton's and Leibniz's introduction of calculus and is, indeed, promoted by Newton himself as the *transient state* that captures the quantum before it disappears, allowing us to treat it as something both *still* existing and *soon* to be no more. As such, it seems to represent Being-Here in a state of Becoming, or both. But Hegel vehemently disagrees with this characterization, saying

> […] that these magnitudes are *either* something *or nothing*; that there is no *intermediary state* between being and nothing ("state" is here an inappropriate, barbaric expression).[28]

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[26] Hegel (2010b, pp. 83–84).

[27] Stekeler (2019, p. 370).

[28] Hegel (2010b, pp. 79–80).

The reason why this example is illustrative of our problem of negativity in logic is, however, more complex as is already evidenced by the fact that Hegel devotes literally dozens of pages of his *Science of Logic* to calculus. I would like to divide this peculiar interest of Hegel in the given piece of mathematics into three interrelated problems:

(1) The problem of *formalism* and its proper evaluation, both with respect to mathematical calculus and its analysis in the categories of logic, such as Being, Becoming, or Being-Here.
(2) The negativity hidden in the concept of the infinitely small quantum. At first, negativity seems to be a means of fully specifying the quantum by negation, in the sense of *determinatio est negatio*, given by the denial of the finiteness of some quantum, but it turns out to be only partial, to be treated as what Hegel calls *bad negation.*
(3) The link between the generality or infinity of some fields of research and their *finitization* by means of negation.

In the remainder of this chapter, I will address all these points, beginning with the *formalism* of the calculus in relation to the formalism of the categories of Being-Here and Becoming. My thesis is that the point of Hegel's critique of infinitesimals does not lie in their logical concept as such, but precisely in the superficiality or formality of their logical treatment: the practices of Newton's and Leibniz's calculus are conceived as dealing with some "fluid" kind of entities—or with motion as such—rather than with some well-construed objects that possess clear criteria of identity and a fixed differentiation between what they "are" and what they "are not."

In this, surprisingly, Hegel's logical analysis is closer to that of Frege—and his interest in calculus as described above—than to that of Marx and Engels, who read calculus directly as *evidence* that the logic of the world, or dialectics, is the "science of the most general laws of *all* motion":[29]

> The turning point in mathematics was Descartes' *variable magnitude.* With that came *motion* and *hence dialectics* in mathematics, and *at once, too, of necessity the differential and integral calculus,* which moreover immediately begins, and which on the whole was completed by Newton and Leibniz, not discovered by them.[30]

Such an idea may arise from the fact that Newton developed calculus in connection with its application in physics, in his *Principia Mathematica*, and conceived the whole matter in temporal terms, as is shown by his terminology of "fluents," "fluxions," "flowing quantities," and so on.[31]

In general, however, calculus was meant to quantify the circumferences, areas, and volumes of complicated or "curved" shapes—such as the orbits of planets—by transforming them into simpler ones, such as squares, rectangles, or triangles. This is

[29] Engels and Marx (1987, p. 545, "Dialectics of Nature").

[30] Engels and Marx (1987, p. 537, "Dialectics of Nature").

[31] Alberto Coffa suggests that it was the Newtonian tradition of founding the calculus in temporal terms that influenced Kant's suspicious idea of making arithmetic dependent on the intuition of time, as opposed to that of space in geometry. See Coffa (1991, pp. 23–24).

something that ancient mathematics already knew, for example, by treating the circle as a polygon with very small—perhaps even infinitely many—sides, but calculus brought a systematic approach to the whole matter, through its own schematic or even mechanical nature and the reduction of multiple problems to two complementary techniques: that of finding the difference, or *differentiation*, and that of summing, or *integration.*

The idea of both, as well as their complementarity, is easily introduced by the picture, see Fig. 11.7. Here, the derivation is to measure the slope of the tangent t to the given curve k at the point P by "taking the difference" in the close neighborhood of P, so that it is expressed as the ratio between the finite but very small differences Δy and Δx. These are the sides of the so-called *characteristic triangle PQR*, which corresponds roughly to the triangle *APB*, of which a part of the tangent forms one side. Since it is obvious that the smaller the triangle, the better the ratio captures the slope of the tangent, the idea arises that the approximation is the best when the differences are as small as possible.

The important point here is that the differences must be both (1) finite, since otherwise they cannot form a difference of $\Delta y/\Delta x$, but would yield a nonsensical expression of 0/0, and (2) less than any finite quantity, since otherwise a better approximation would be possible. A similar consideration applies in the case of integration, where the area under the curve k bounded by the x-axis is assessed. In this case, the area is divided by a finite number of rectangles, with the same observation that the smaller these rectangles are, the better the approximation of the area.

The question is in what sense one does the summation over an infinite number of infinitely small quantities, and in what sense these are quantities at all.

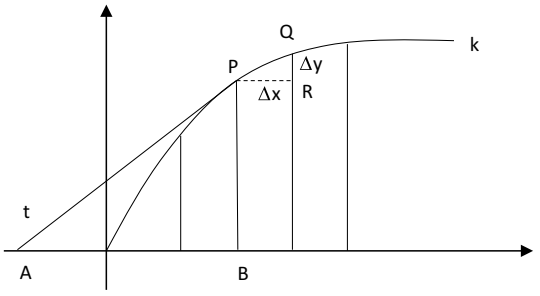


Fig. 11.7 Differentiation and integration

11.7 Bad Negation

What Engels does in his *Anti-Dühring* and later in the *Dialectics of Nature*—following some remarks by Marx concerning on the significance of calculus, which were later pretentiously collected under the title Marx's *Mathematical Works*—is in fact to ignore the whole problem. Although he repeatedly refers to "the old Hegel" who is usually "quite right," even on matters of foundations of calculus,[32] he *seems* to follow only—that is, formally—the dialectical method of two denials, first in antithesis and then in synthesis, relying on the traditional fear of contradiction:

> I differentiate x and y, i.e., I take x and y as so infinitely small that in comparison with any real quantity, however small, they disappear, that nothing is left of x and y but their reciprocal relation without any, so to speak, material basis, a quantitative ratio in which there is no quantity. Therefore, dy/dx, the ratio between the differentials of x and y, is equal to 0/0 but 0/0 taken as the expression of y/x. I only mention in passing that this ratio between two quantities which have disappeared, caught at the moment of their disappearance, is a contradiction; however, it cannot disturb us any more than it has disturbed the whole of mathematics for almost two hundred years. And now, what have I done but negate x and y, though not in such a way that I need not bother about them any more, not in the way that metaphysics negates, but in the way that corresponds with the facts of the case? In place of x and y, therefore, I have their negation, dx and dy, in the formulas or equations before me. I continue then to operate with these formulas, treating dx and dy as quantities which are real, though subject to certain exceptional laws, and at a certain point *I negate the negation,* i.e., I integrate the differential formula, and in place of dx and dy again get the real quantities x and y, and am then not where I was at the beginning, but by using this method I have solved the problem on which ordinary geometry and algebra might perhaps have broken their jaws in vain.[33]

I quote this to emphasize how Engels' attitude differs from that of Hegel's *Science of Logic*, not only on the question of infinitesimals, but also on difference between *Becoming* and *Being*. In Hegel's view, this type of analysis is not a product of the dialectical method, but rather a lazy work that *merely* denies that something is quantifiable. As such, it is not capable of guaranteeing a new quantum, or Being-Here, but rather some transient state which, as he says above, is not a state at all:

> The infinitely small signifies in the first instance the negation of the quantum as such, that is, of the so-called *finite* expression, of the completed determinateness that quantum as such possesses.[34]

Conceptually, of course, this is brand new information that strikes at the idea of *determination* as *negation*, showing that "mere" denying is not enough, or only leads to what is called a *bad negation*, as opposed to the "true" or "determinate" one. As Stekeler points out here, however, the word "bad" here is not meant as a moral judgement or an evaluation of intrinsic quality, but rather as a mistranslation of the

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[32] Marx (1983, p. XXVII, Engels' letter to Marx dated August 10, 1881).

[33] Engels and Marx (1987, pp. 127–128, "Anti-Dühring").

[34] Hegel (2010b, p. 258).

German "schlecht" in its original meaning of "schlicht" or "mere," which has been preserved in collocations such as "schlechtweg" or "schlechthin."[35]

This makes the point about infinitesimals and the way they were introduced quite general, since they obviously did not come out of nothing—or out of Pure Being in its original, unqualified nature—but against our ability to assign numbers to given magnitudes, whether rational or even real. In other words, there were already some differences, along with our ambition to introduce new ones by the *mere negating* of those original differences on the grounds that they are not specific enough. This is what the process of approximation—and the category of *Becoming* behind it—represents. The infinitesimals thus correspond to:

(1) the idea of obtaining something *between* the approximation and its result, a new kind of quantity, which must be *infinitely small*;
(2) the idea of what the approximation seeks to approximate, which, being determined by an infinite series, must be *infinitely large.*

Both of these ideas are, of course, confusing, having appeared long before a thorough conceptual clarification of what they might mean was given in the work of Cantor. They are related to, if not identical with, the classical paradoxes of Zeno, such as that of the flying arrow in the case of the infinitely small, and that of Achilles and the tortoise in the case of the infinitely large.

Calculus incorporated these paradoxes, as based on the ideas (1) and (2), into its practice, which, to the despair of both Newton and Leibniz, included both the possibility of dividing by dx, making it different from 0, and the possibility of annihilating the factor in which dx occurs, making it equal to 0. The excuse was that it (sometimes) works. The result is *formalism* at work, not the victory of contradiction, as Engels thinks. And it is this formalism that Hegel ridicules when he speaks of the whole procedure (repeating Berkeley without quoting him) as a *compensation of errors*.[36]

What we are allowed to do, Hegel claims, is not to assign to these infinitesimals the status of Being-Here, but rather that of Becoming, which is precisely the *relationship* between these approximations, or rather the infinite series of them, and the result:

> The series exhibits the contradiction inherent in it of representing something which is a relation, and possesses *qualitative* nature within it, as without relation, as a mere *quantum*, as amount. The consequence is that there is something always lacking to the amount expressed in the series, so that the posited amount must always be gone beyond in order for the required determinateness to be attained.[37]

This is not to say that I cannot transform the Becoming employed here into Being-Here, that is, transform an implicit relationality into an explicit object. Just recall the

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[35] Stekeler (2005, p. 244).

[36] Hegel (2010b, pp. 226–227). Berkeley famously analyzed Newton's method of calculus as being misleading and referred to the concept of infinitesimals as "the ghosts of departed quantities," noting that calculus works precisely because of compensating errors. See Berkeley (1992, pp. 184 and 199, "The Analyst").

[37] Hegel (2010b, p. 210).

case of relational—or unsaturated—expressions in Frege and their transformation into saturated ones, that is, names (Sect. 10.6). But to do this, it is not enough to assert that this or that object exists. One must *draw* a boundary and do so *in a way* that takes into account other boundaries by relating it to them—in this case, by relating infinitesimals to the objects already established, such as natural and rational numbers. This is how I read statements such as:

> […] since the infinitely great or small is such that it can no longer be increased or diminished, it is no longer in fact a *quantum* as such.[38]

Without having established standard relations such as $<$, $+$, and, most importantly, $=$, one cannot know what the status of infinitely great or small is. This is exactly what Frege says about the directions of straight lines, insisting that one must first establish the identity criterion and then settle all the remaining relations as given by their arithmetical context.

Thus, the *true negation*, as opposed to the *bad one*, is not a specific kind of denial, but rather a denial that—like identity, which denies the difference of some phenomena—embeds the given negation in the network of already established negations, connecting it with the totality of distinctions and the *generality* that derives from them.

11.8 Bad Infinity

Being conceived as the *bad result* of the process of approximation, in the sense of the unstable quality of our knowledge, makes the analysis of infinitesimals relevant to general matters of logic, particularly with regard to the link between negation and universality. Here the logical concept of "bad infinity" appears as a term for conceptual *indeterminacy*, that is, for the possibility of denying all established distinctions and crossing all drawn lines in favor of some transcendent result:

> Only the bad infinite is the *beyond*, since it is *only* the negation of the finite posited as *real* and, as such, it is abstract first negation; thus determined *only* as negative, it does not have the affirmation of *existence* in it; held fast only as something negative, it *ought not to be there*, it ought to be unattainable. However, to be thus unattainable is not its grandeur but rather its defect, which is at bottom the result of holding fast to the finite as such, *as existent*. It is the untrue which is the unattainable, and what must be recognized is that such an infinite is the untrue.[39]

Such an untrue or bad result, which does not deserve to be called Being-Here or real, contributes to the range of Becoming, beginning with the original indeterminacy, represented by Pure Being, which is infinite in the sense of many arbitrary directions in which one can limit oneself, each formally as good as the other; or in the idea of denying the differences established so far, either because one can or because

[38] Hegel (2010b, p. 206).

[39] Hegel (2010b, p. 119).

something forces one to do so, without giving any indication of which direction to take. The latter is what Kuhn called the opportunity for a *paradigm shift* or scientific revolution.

What Hegel can agree with Kuhn on is that such an opportunity for shift is always present, in an abstract sense, because our determinations are affected by their inherent negativity and thus never fully "in accord with reality," as correspondence theory and fallibilism like Popper's would have it. But the idea of achieving what is real by simply throwing away these differences is irrational, if not insane, as the concept of "bad infinity" attests. In its simplest form, as described in the quote above, it amounts to the observation that our distinctions are inexact, or vague, promising more than they deliver, leading to a mere negation of them in the idea of the true world in which they are what they are, similar to the ideas of Platonic heaven.

Bad infinity, in this sense, is another term for *alienation*, now described as the belief that one can arrive at something determinate simply by denying previous determinations. The result, however, is only a step into the infinite field of possibilities without the support of the preceding series of decisions, no matter how fragile they may be. What Hegel does in his exegesis of calculus is to provide a model of what bad infinity can mean in the logical sense of the word, in terms of the relationship between approximation processes and their results. His favorite example is that of the decimal expansion 0,285714… This is initially only negatively determined, in that it is always expandable, and thus does not lead to any entity such as a rational or real number.[40]

In Hegel's choreography, this deficiency can be remedied by negating the given indeterminacy with a *second negation* found in the subsequent finite specification of what the sequence is aiming at—in this case, the ratio $2/7 = 0{,}285714\ldots$ As noted above, this step must embed the prospective object into the realm of the previously established rational numbers as an alternative mode of their description, transforming a "mere" Becoming into Being-Here. This embedding is, of course, a problem in itself, as was manifested in the case of infinitesimals, and must involve the task of showing how the infinity presented here, which is both the "infinitely large" of the increasing series of rational approximations 0,2; 0,28; 0,285; 0,2857… and the "infinitely small" of the difference between this series and its result, can be treated in an orderly fashion.

Both of these subtasks were handled by Cauchy's reform, in which the idea of the rational convergence of a series to a value—conceived as consisting in an *infinitely small* difference between them—was rephrased so that there was no need to use this problematic reference at all:

> A series $(a_i)_{i\in\mathrm{N}}$ of (rational) values converges to the value of b if and only if, for any given (rational) limit of c, there exists an index i from natural numbers N such that the difference between b and all a_j beyond a_i (that is, with $j \geq i$) lies within c.[41]

[40] Hegel (2010b, pp. 209–211).

[41] The notation $(a_i)_{i\in\mathrm{N}}$ means that the values a_i are indexed by the natural numbers N, as in a_1, a_2, a_3..., but N here, in fact, can actually stand for any linearly ordered set, or set of points to be identified beforehand on the abscissa (x-coordinate of the figure).

See Fig. 11.8. Frege's analysis of the calculus allowed for the precise articulation of this decision to avoid the substantive need for infinitesimals, making them mere *syncategorematic* parts of the whole expressive enterprise. Even in this non-substantive role, however, they can cause more confusion than benefit if not handled wisely. And this is the *illusion* that, in the case of convergence, the fact that for every c there is some point a_j from which the difference between the following members and b is less than c means that there is one fixed value of c for all points, which would make this difference infinitely small.

What Cauchy's reconceptualization did was not to make the bad infinity of the infinitesimally small *good*, in the sense of transforming infinitesimals into the Being-Here (which was done much later in the non-standard analysis of Abraham Robinson), but to render them *non-existent*, in the sense of eliminating them—or their substantial use—from discourse. However, this does not apply to the concurrent concept of the "infinitesimally large," first exemplified by the decimal expansion of 0,285714..., which is now determined by its equation with 2/7, not arbitrarily, but by establishing the conditions under which this equation holds via the previously clarified concept of *convergence.*

Our interest in these examples as a means of turning bad infinity into good, in the sense of overcoming the alienation inherent in them, becomes more transparent when we consider cases in which decimal expansions do not have such an easy "finite summation," precisely because they do not approximate a previously defined rational number. In Hegel's time there was not yet an official arithmetical way of dealing with this, which would come only a few years later in the work of Georg Cantor—along with the possibility of introducing the "infinitely large" as an object in the more ambitious sense of the word. But there was also an ancient precedent for the discovery that such unlimited, yet finite, expressions exist, namely in the *irrationality of numbers.*

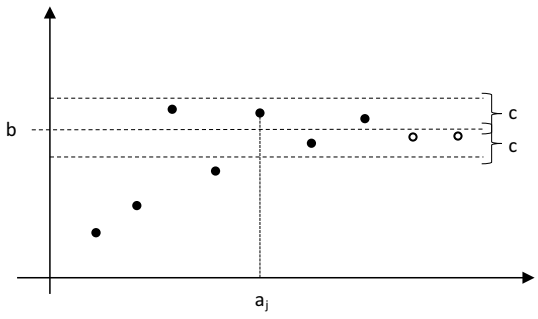


Fig. 11.8 Transcendent convergence

11.9 Finite Infinity

It is the phenomenon of *irrationality* associated with the rejection of traditional modes of quantification that represents the case of bad infinity transformed—and triumphantly so—into good infinity in the concept of the *real* number. As such, it stands for the *finitization of infinity*, which can be characterized as the process of making the infinite possibilities of determination immanent without losing their inherent tendency to transcend the finiteness of mere fixed determinations, or their tendency to be "finite" in a "bad" sense corresponding to the "badness" of their simple denial.

This logical point can, in fact, be articulated within the threefold structure of the *dialectical* method, beginning with what might be called a *thesis* of "rational" coping with the world:

(T) Every magnitude, including sets, lengths, areas, or volumes, is determinable in an orderly fashion, or measurable.

Such measurement, generally, consists in giving a *determinate* answer to the question "how many times does magnitude B, treated as a unit of measure, fit into the measured magnitude A?" This answer is quite straightforward in the case of discrete magnitudes, such as sets, which by definition consist of units, and more complicated in the case of continuous magnitudes, such as lengths or volumes, where such a presupposition is spurious.

The idea expressed in (T), though, is that both cases, discrete and continuous, can be handled similarly. In the latter case it is complicated by the phenomenon of the remainder R, to which the answer "how many times B fits into A" usually leads. Put another way: it is different to count chickens, where one chicken is the unit and leads to the definitive answer *n*, than it is to measure the length of a rope, where after subtracting the given unit—such as foot or cubit—*n* times, something is usually left over that is, by definition, smaller than the unit. One way of dealing with this, which we now practice, is to divide the given measure (our "meter") B into a smaller number of parts—such as 10 in the decimal system, analogous to the "inch" of the original "foot." With this "new measure," the remainder R, if any, from the previous procedure is repeatedly measured, resulting in what we know as decimal expansion and the eventual common divisor between A and B.[42]

The ancient method of measurement is more elegant than this: instead of dissecting the measure relatively to a previously chosen number, it simply works with what it already has, switching the sides of what is being measured and what serves as the measure, making the remainder R the new measure of the old unit of measurement. This is why it is called *anthyphairesis*, or "reciprocal subtraction." When applied to

[42] The problem here, known from decimal expansions such as 0,285714..., is that these expansions often remain infinite even though they correspond to a rational number such as 2/7; in other words, there is a common divisor, but it is not the one corresponding to the division of choice. However, these infinite expansions of rational numbers are the ones that eventually become periodic, from which a rational representation in terms of two natural numbers *p*/*q* can be constructed. But that is another topic.

discrete quantities, such as sets, it is commonly used to find their *greatest common divisor*—as in the so-called Eudoxos algorithms—which is the last remainder before the procedure stops. For obvious reasons, this works for numbers, and should also work for continuous quantities if there is a common divisor at all.

What happened, unexpectedly, in the mutual measurement of magnitudes within some basic forms, such as the pentagon or the square, was the discovery that such a divisor is missing, thus providing the antithesis (A) to (T):

(A) There are magnitudes that cannot be measured in an orderly way.

The case of the pentagon is particularly illustrative. Based on the fact that each side of the pentagon is parallel to some of its diagonals, the following relations can be established: D = S + A and S = B + A. See Fig. 11.9. The reciprocal subtraction of S from D, then, gives, in the first steps, A and B, which, however, are the side and the diagonal of the inscribed pentagon. So the whole process must go on infinitely, without giving any measure or its expression, such as 2/7.

In ancient mathematics, this finding was interpreted as the collapse of the old theory of proportions, which held that proportions could be rationally—or logically—expressible by two natural numbers. As a consequence, proportions such as that between the diagonal and the side of a pentagon are simply "bad," leaving us with an infinite measuring process, or Becoming, instead of the result to which it is supposed to lead in the original sense of (T). The positive side of this was extracted by Plato, who rightly saw in these new "bad" proportions the refutation of the simplistic finite position, which is also "bad," simply because it treats the established determinations as given.[43] The *irrational ratios*—or *alogoi logoi*, as they were called—represented something that was not empirically demonstrable, since all empirical measurements yield a finite—that is, rational—result. In fact, they represent a rational ground for

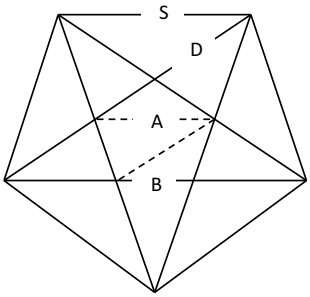


Fig. 11.9 Incommensurability

[43] See, for example, passages such as: "The real relationship between commensurables and incommensurables. We must be very poor specimens if on inspection we can't tell them apart. These are the problems we ought to keep on putting up to each other, in a competitive spirit, when we've sufficient time to do them justice; and it's a much more civilized pastime for old men than draughts" (Plato, 1970, pp. 314–315/820).

transcending the mere finiteness of this world as expressed in the original formulation of (T).

Since the infinite alternative (A) was not particularly "good" either, the original theory was replaced by the more geometric approach captured by Eudoxos' theory of proportions. At the same time, however, it was obvious to many that the real situation of (A), at least in practical terms, was not as "bad" or "undetermined" as it might originally appear, since the resulting sequence of numbers corresponding to the answer to how many times B fits into A was not entirely arbitrary. In the case of the pentagon, it is even determined as it goes, having the very simple "ratio" 1, 1, 1, 1...

These seeds of goodness are not, of course, sufficient to transform the whole process into a result, or to specify what such a result—the Being-Here of the *alogoi logoi*—might be, as in the case of sequences that approximate some existing number, such as 2/7, which, in the *antiphairetic* case, are precisely those that are finite. What does it mean, for example, to multiply such an *alogoi logoi* by another, or to increase it by 1? At this moment, what we have is nothing more than an *external reference point*, or the Beyond of the approximation process, of which, not unlike Popper's verisimilitude, we only know that we *want to have* it in order to perceive the procedure as capturing the result of cognitive progress. And since there seems to be no "good" immanent way to get that result, it makes no sense to say that the series *converges* to it or is *progressive* at all.

In mathematics, this manifests itself in the confusion between what are now called divergent series—or non-progressive ones—, which in Euler's attempts to treat them as converging in infinite sums such as $1 - 1 + 1 - 1 + 1...$, led to different results: 1, if read as $1 + (-1 + 1) + ...$, and 0, if read as $(1 - 1) + (1 - 1) + ...$ The same goes for Leibniz's idea of considering the arithmetic mean of 1/2 for such a result. The idea of establishing the convergence—or the goodness—of series by means other than transcendent ones first came from the infinitesimal approach, by claiming that the difference between its members tends to be *infinitesimal*—a notion that, as we know, presupposed a vague notion of infinity and caused another inconsistency, as described above. Fortunately, once the concept of infinitesimals was corrected in

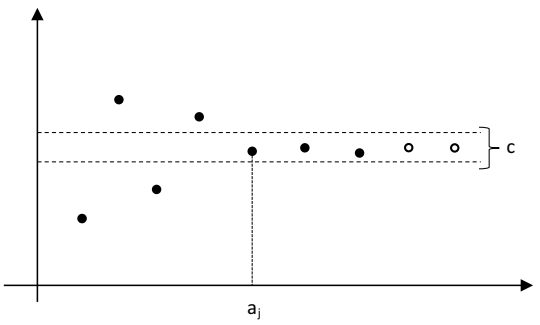


Fig. 11.10 Immanent convergence

Cauchy's manner, as Cantor later did, it led to a workable criterion of convergence by immanent means:

> A series $(a_i)_{i\in\mathrm{N}}$ of (rational) values converges (not to some value, but *per se*) if and only if, for any given (rational) limit c, there exists an index i from natural numbers N such that the difference between all a_j beyond a_i (that is, with $j \geq i$) lies within c.

See Fig. 11.10. This is to say that from *some point* we can squeeze the members of the series within any limit we want, no matter how small it is, with the Fregean remark that this does not mean that there is some point for every limit, but rather the reverse, that for every limit there is some point. This led Cantor to the definition of a real number (whether rational or irrational) as a series of rational numbers that satisfies the given immanent criterion of convergence without the need to refer to some previously given point of it, as described in the definition above (Sect. 11.8).

Conceptually, this captures the most important insight, according to which the given progression, as the approximation of something—be it a number, a point, or a state of knowledge—does not lead to some pre-existing point of reference, but can be treated *as that very point of reference itself* once the given conditions of convergence are met! As such, it provides what might be called a synthesis of the previous points (T) and (A), which takes us back to the earlier claim that everything *is* measurable *but* by a different kind of means, whereby the process of the old measurement, or the prior Becoming of the infinite approximation, can serve as its result, namely the finite determination, or finitization, of infinity:

(S) Everything is measurable in an orderly way, which combines the finiteness of the former method of (T) with its infinite extension as given by (T)'s denial (A).

Metaphorically speaking, the journey became its own goal and the process of knowledge its own result. As a way of transforming the Becoming into the Being-Here, however, this journey is anything but straightforward. It does not consist in the simple denial of something, but in the further determination of this denial—or, let us say, in denial of this other denial—by means such as Cauchy's criterion of (immanent) convergence.

11.10 Existence and Reality

In light of the previous section, the concept of Finite Infinity—as a combination of the logical concepts of Universality and Negativity—introduces two crucial insights into the entire logical edifice, which must be considered together.

(1) One is the *dialectical structure* that further diversifies and complicates the original idea of the determination of Pure Being by means of the infinite series of negations that are supposed to lead to Absolute Being as the Pure Being's closure.

(2) The other is the critique of the mere formality of Being in its opposition to Becoming.

The case of *real* numbers—not named as such by chance—makes this complicated structure explicit, in a logical model that exemplifies both these insights.

On the one hand, there are fixed determinations, measurements by the old standard, in the model represented by natural and rational numbers. These numbers stand for the established Being-Here, which, however, can easily become empty or alienated if treated as existing on its own, divorced from the process that led to it, the process of measuring. This divorce is what the concept of *Bad Finiteness* (T) can stand for, an arbitrary theory separated from its history, or the cognitive process, which is captured by the category of Becoming. This category motivates, in the next step, to find the established determinations insufficient or to reject them if they fail to "measure" some relatively common magnitudes. In an ironic twist, then, another "bad" attitude can arise if this failure is treated as a direct approval of the existing state, for example, by naming the convergence point as lying there, in the infinity of the determinations' series, or as having an infinitesimal difference from it.

Such a position is only a nominal option that unwillingly sacrifices the concept of Being and Being-Here for the existence of Becoming, thinking of it as the true kind of Being instead. Hegel, unlike his followers such as Marx and Engels, decisively rejects this idea, insisting that Becoming, rather than a new kind of "transient" Being, is the *Relation* between the Being employed in the process of self-determination, as represented by the series of rational numbers, and its prospective result. What is to be done, then, is to demonstrate how the old determinations can function as a new kind of measure, that is, to reconcile Becoming, as the ever-present process of experience, with Being-Here, as the result to which this process can ultimately lead. Otherwise, one is faced with a different kind of alienation, where *Bad Finiteness* is supplemented by *Bad Infinity* (A).

Against this background, *the Finite Infinity* (S)—and the dialectical process behind it—provides more than a mere dialectization of concepts, in the sense of making them dynamic, but also the *radical critique* of reality's formal treatment, as captured in the formalism of Being, in its formal delimitation from Becoming. Such a critique, then, concerns the dialectical method itself, if it is read as a new formalism rather than as something that such formalism both presupposes and nullifies. In this critique, the logical concepts of Negation and Universality operate in an expressively and epistemologically adequate way that is precisely suited to the context of immanent and fallible experience. In doing so, they oscillate between the concepts of *bad immanence* and *bad transcendence*, which must be balanced by mutual coordination in order to avoid alienation.

Plato, for example, even in his later epistemology, remained somewhere on the edge of both, between his metaphysical stance, in which ideas reside in a transcendent realm of intelligibility, and the immanent means of reaching them, inspired by the phenomenon of *anthyphairesis* (Sect. 11.9) in its logical projection onto the method of *diairesis*, or the division of concepts. In this latter method, the identity of meaning was to be achieved in a dialogical process of arriving at a "common divisor" of

opinions by means of a series of determinations D_1, D_2, D_3, D_4..., the sum of which yields the concept in question.[44] In all this, while sharing Plato's starting points, Aristotle tended toward the opposite side of the dilemma, or toward Bad Finiteness.

In contrast to both Plato and Aristotle, Wittgenstein attempts to reconcile this finite infinity of our experience by approaching the concept of A historically, through the sequence A_1, A_2, A_3, A_4... of A's introduction. In its finiteness, he argues, this sequence *cannot* suffice, since it can be extended in many different directions; but paradoxically, he adds, it *can*, of which the concept of A—treated as a *rule* one is able to follow—is an expression. As rules that can be followed, not just once, but an unlimited number of times, the concepts are examples of a finitized infinity that is supposed to overcome the previously established paradox in a *skeptical* way. (See Sect. 5.5.) In the end, however, Wittgenstein's skeptical solution is still an example of the "bad" infinite thinking, since no such thing as an infinite rule is possible, either because all our finite distinctions are always too inexact or open to modification or because true exactness is infinite and thus unattainable.

The true solution to the paradox, based on the logical categories of Hegel, can now be articulated as follows: There is the *infinity* of Pure Being, in our ability to further stratify this Being in an indeterminate number of ways, and the *finiteness* of the Being's stratification that we have achieved so far, in the process of knowledge. These stratifications are, in an act of bad negation, easily seen to be arbitrary and thus infinite too, extendable in multiple directions. But such a possibility is only *theoretical*, limited precisely by the *historicity* of the introduction of these stratifications by the reasons we have for them, and by the very fact that we rely heavily on their existence even when we are doubting them. The logic of negation takes all this into account and treats everything we have as a combination of the infinity of possibilities with their finite substantiation, or what might be seen as the determination of generality in a stratified way in which the negations employed cooperate to form the conceptual system, or what Hegel calls the *Concept* (*Begriff*). See Fig. 11.11.

Thus, for example, the introduction of the concept A, such as that of a cat, depends on our individual history of the infinite negations of $A_1 = A_2 = A_3 = A_4$..., but also on the connection of these negations with other concept(s) B, such as a dog, which we know is not a cat, that is "$A \neq B$," even though in individual cases one may be entitled to refer to both, for example, when introducing the concept C, such as a the

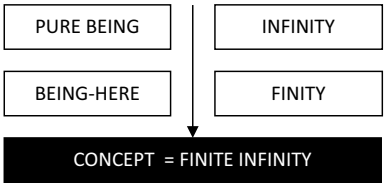


Fig. 11.11 Finity infinity

[44] On the parallelism of *anthyphairetic* definition and the method of *diairesis*, see, for example, Stekeler (1995, pp. 80–82).

mammal. Similarly, and in a more complex way, in the case of the real numbers, it is not only the "convergence" criterion that makes them representatives of what finite infinity means but the ability of this criterion to embed them in the original contexts of measurement, establishing relations with original entities and operations, beginning with that of identity.[45] This provides convergence in a broader or *proper* sense, namely the *convergence of knowledge*. This is why, in the following chapters, the real numbers will be able to represent both a model of certain logical aspects of knowledge and an example of it. See Fig. 11.12.

This brings us to another aspect of Finite Infinity, which takes us from the dialectical level to the final, speculative one, and that is the fact that not only the newly introduced entities, cats and reals, are infinite in a sense, because they are based on the infinite series of the old determinations A_1, A_2, A_3, A_4... or 3,14159..., now finitized by a rule to be followed. What is infinite in a primary sense is the relation between these new entities, such as cats and reals, and the old determinations, which are fixed and finite, in the possibilities of their extension. As such, this infinity of theirs not only overcomes the mere formality of logic in a dialectical way, but also establishes the new, speculative meaning of logical concepts with respect to the whole thus established.

It is precisely in this sense that the true difference between the mere *Existence* of some determination and its *Reality* (*Wirklichkeit*) appears. The latter stands for something more than the mere possibility of drawing a boundary in this or that way, but for drawing it for a *reason*, similar to the one we apply when we treat series of rational numbers as new, real numbers, or when we treat pictures in terms of linear projection rather than as they merely appear to us. It is such a reason that, as Hegel famously says, makes the real rational (and vice versa) and renders truth as the matter of the whole.

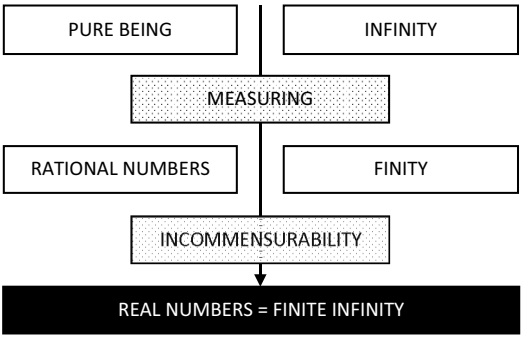


Fig. 11.12 Real numbers

[45] This is also defined by the Cauchy criterion, which is based on the differences between the members of the given number series. Two Cauchy series represent the same number if and only if for every limit, no matter how small, there exists a point after which the differences between the members of both series are smaller than this limit. The definitions of operations such as + and × follow in a similar way.

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